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西北工业大学研究生学位论文选题报告表

Northwestern Polytechnical University
Graduate Degree Dissertation Title Selection Report Form

学　　院 School	计算机学院 School of Computer Science
学 科、 专 业 Discipline / Specialty	计算机科学与技术 Computer Science and Technology
姓　　名 Name	大雷 DAULET ANEKESH
学 位 级 别 Degree	硕士 Master
导　　师 Supervisor	郭斌 Prof. Bin Guo
培 养 类 别 Category	留学研究生 Overseas Postgraduates
报 告 日 期 Date of Report	2026年10月 October 2026

研　究　生　院

Graduate School

研究生学位论文选题报告的要求

一、 硕士生的选题报告内容应包括文献综述、选题意义、研究内容、研究方案，论文工作量的估计、工作条件，预期达到的水平，存在的问题及拟采取的解决措施。

二、 博士生的选题报告内容应包括文献综述、选题背景及其意义、研究内容、研究特色、工作难点、预期成果及其可能的创新点。

三、 选题报告会应以学术活动的方式公开进行。

四、 正式开题之前，研究生应在广泛阅读中、外文资料的基础上，深入了解拟选课题的国内外研究动态，把握所选课题的目的、意义和预期结果，明确课题工作的设想、方法和研究路径。

五、 研究生在规定的时间内，写出选题报告初稿，经指导教师审阅同意后，由指导教师安排选题报告时间。选题报告未通过者，重新开题，若第二次选题报告仍通不过者，则按有关规定终止学籍。

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Requirements for the Dissertation Title Selection Report

1. The Dissertation Title Selection Report for Master Degrees should include the Literature Review, the Meaning of the Title Selection, the Research Contents, the Research Plan, Workload Estimation, Working Conditions, the Goal to Achieve, Problems Existing as well as the Measures to Take.
2. The Dissertation Title Selection Report for PhD Degrees should include Literature Review, Background and Meaning of the Selection, Research Contents, Characteristics of the Research, Difficulties, Goals to Achieve and possible Innovations.
3. The Report should be made in public as academic activities.
4. Before starting the title, graduates should, based on the extensive reading through the literature both in Chinese and in foreign languages, have a deep command of the research tendency home and abroad. They must have a clear mind of the aim, meaning and foreseen results, research assumption, methods and approaches.
5. Graduates should finish the draft report within the required time, and then the supervisor after having checked the report, organizes time to make the report. Those who fail to pass the report have to reopen a title. If it can not be passed for a second time, the studentship will be stopped.
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7. The form should be filled out in pen or printed. If the pages are not enough, extra pages can be attached.

论文题目 **Dissertation Title**	**Morphologically-Grounded AI-Generated Text Detection and Factual Verification for Low-Resource Agglutinative Languages: A Comprehensive Framework for Kazakh**			
论文类型 （在有关项目下作记号√） **Dissertation Type** (Put a tick √ under relevant item)	基础研究 Fundamental Research	应用研究 Application Research √	工程技术 Engineering Technology	跨学科研究 Research Cross Discipline
研究生在进行选题报告、听取意见后，整理成文（不少于 3500 字）。 **After the report and comments, make the records up to an article. (no less than 3500 words)**				

1 Research Background

1.1 Relevance and Context

The rapid evolution and widespread deployment of autoregressive Large Language Models (LLMs)—such as OpenAI's GPT-4, Meta's LLaMA-3, DeepSeek-R1, and multilingual open foundation architectures including Cohere's Aya—have fundamentally transformed artificial intelligence, automated reasoning, and natural language communication [1–5]. These foundation models demonstrate exceptional proficiency in generating syntactically coherent, stylistically diverse, and contextually fluent texts across numerous domain contexts. However, the universal accessibility of advanced generative models has simultaneously escalated severe risks concerning information integrity, epistemic credibility, and digital safety [6, 7]. Contemporary digital ecosystems face increasing exposure to synthetic text artifacts, ranging from automated disinformation campaigns and fraudulent consumer reviews to AI-assisted academic plagiarism and hallucinated pseudo-factual assertions delivered with unmerited confidence.

Consequently, establishing robust computational mechanisms for auditing digital text content has emerged as an imperative requirement for trustworthy artificial intelligence. Reliable information verification demands two foundational capabilities: (1) *AI-generated text detection*, which identifies generative origin by discerning whether a candidate text was composed by a human or synthesized by an autoregressive neural network; and (2) *automated fact verification*, which evaluates whether factual propositions are substantiated by reputable reference evidence.

While extensive research has advanced text auditing in high-resource, typologically analytic or mildly fusional languages such as English and Standard Mandarin, existing methods encounter fundamental performance bottlenecks when deployed in low-resource, morphologically complex agglutinative languages [8, 9]. Languages of the Turkic family—exemplified by Kazakh, which serves as the official state language of Kazakhstan and is spoken by over 15 million people across Central Asia—possess typological structures that directly challenge standard deep learning assumptions. In agglutinative linguistic systems, lexical roots serve as anchors for extensive, sequential suffixation chains that encode negation, syntactic case relations, tense, evidentiality, possessive markers, and pragmatic moods within single polymorphemic words. A solitary verbal root can produce hundreds of distinct surface wordforms without introducing new lexical headwords.

The proliferation of multilingual generative models within Central Asian educational, administrative, and public communication ecosystems underscores an urgent sovereign need for language-native text verification infrastructure. Without specialized auditing tools tailored to the morphological prop-

erties of Kazakh, regional media environments remain acutely vulnerable to synthetic manipulation. Developing linguistically grounded detection and verification frameworks is therefore essential for ensuring information security and linguistic equity.

1.2 Challenges of Existing AI Detection Systems

Despite substantial progress in machine-generated text detection, contemporary detection paradigms exhibit critical vulnerabilities when applied to Kazakh and typologically similar agglutinative languages:

(1) **Subword Tokenization Fragmentation:** Prevailing transformer architectures rely on statistical subword algorithms, such as Byte-Pair Encoding (BPE), WordPiece, or SentencePiece [8]. These algorithms are predominantly trained on high-resource multilingual corpora, optimizing vocabulary merge operations based purely on surface frequency rather than grammatical structure. In agglutinative languages, standard tokenizers aggressively partition polymorphemic words across arbitrary character boundaries, fragmenting functional affixes into meaningless byte sequences. As illustrated in Figure 1, the Kazakh word *мектебіміздегілерден* ("from those who are in our school") contains a transparent grammatical sequence of root, possessive, locative, adjectivizer, plural, and ablative morphemes. Standard subword tokenizers partition this single word into disjoint subword fragments, obscuring both the root lemma and the inflectional markers. This fragmentation inflates effective sequence lengths, depletes transformer context windows, and attenuates gradient propagation during representation learning.

(2) **Absence of Morphological Priors and Linguistic Grounding:** Conventional neural detectors treat text as unconstrained surface token sequences, lacking explicit inductive biases regarding root-affix compositionality, vowel harmony (palatal and labial harmony), and morphotactic suffix ordering [10, 11]. As a result, neural classifiers struggle to distinguish natural inflectional richness from synthetic generation artifacts, frequently misclassifying formal, highly inflected human prose as machine-generated text.

(3) **Severe Cross-Generator and Out-of-Domain Degradation:** Zero-shot probability curvature methods (such as DetectGPT [12], Binoculars [13], and Fast-DetectGPT [14]) depend on surrogate language models to evaluate local log-likelihood curvatures. In low-resource agglutinative languages, available surrogate models possess undertrained parametric vocabularies, resulting in unstable log-probability surfaces that destroy the detection signal. Supervised neural classifiers, meanwhile, overfit to generator-specific surface regularities [15, 16]. When evaluated across unseen foundation model architectures (e.g., training on Sherkala-7B and testing on Qwen-2.5-7B-Instruct) or cross-domain distributions (e.g., news versus customer reviews), standard detectors experience catastrophic drops exceeding 30 percentage points in AUROC.

(4) **Inability to Handle Localized Synthetic Tampering:** Conventional document detectors employ uniform sequence truncation (e.g., the first 512 tokens) or naive average pooling, assuming that documents are entirely human-written or entirely synthetic. In real-world dissemination, deceptive texts are frequently created by splicing isolated synthetic sentences into authentic human articles. Naive global pooling dilutes localized forensic signals, leaving standard architectures incapable of detecting boundary-spanning hybrid manipulations.

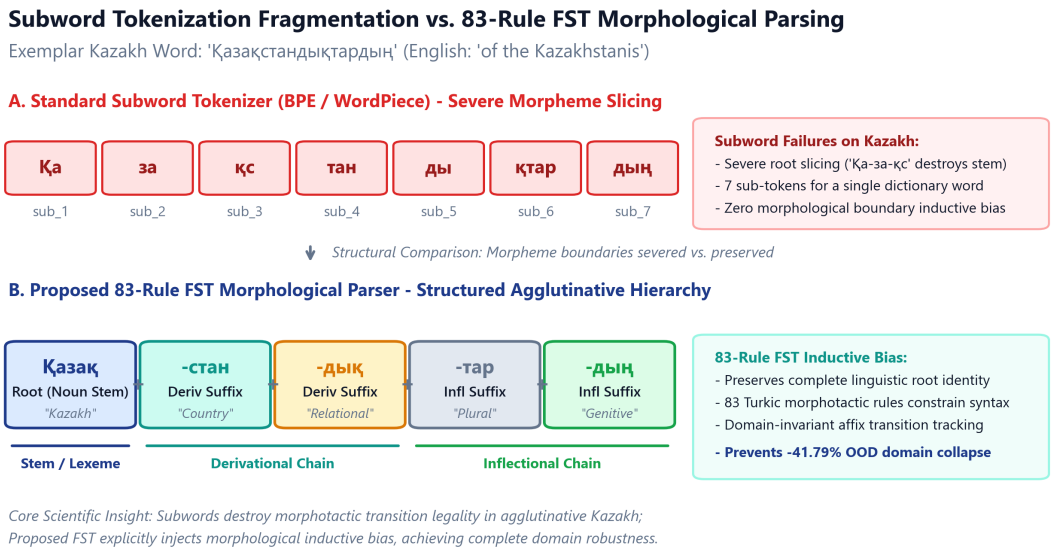


Figure 1: Subword Tokenization Fragmentation vs. 83-Rule FST Morphological Parsing in Kazakh.

1.3 Potential of Morphologically-Grounded AI and Verification

To overcome the architectural and typological limitations of existing detection systems, this research proposes a morphologically-grounded computational framework that unifies symbolic morphological modeling, deep transformer representations, and multi-stage factual verification.

First, integrating formal computational morphology via an 83-rule Finite-State Transducer (FST) provides precise, bidirectional decomposition of complex agglutinative words into their canonical roots and structured inflectional tags [10, 11]. Unlike statistical subword tokenizers that fragment words arbitrarily, the 83-rule FST explicitly models Kazakh vowel harmony, consonant assimilation, and morphotactic constraints. By coupling this symbolic morphological analyzer with pretrained deep contextual representations (such as Kaz-RoBERTa [17]), our framework establishes a dual-stream architecture wherein symbolic linguistic priors directly guide neural feature extraction. This hybrid formulation neutralizes subword fragmentation, stabilizes gradient propagation, and substantially bolsters cross-generator generalization.

Second, this research recognizes that comprehensive text safety requires decoupling *generative origin* from *factual veracity*. In real-world media environments, machine-generated texts can articulate verified truths (e.g., automated sports or meteorological summaries), while human-authored text may propagate deliberate disinformation or rumors. Conversely, synthetic prose can hallucinate falsehoods, and human writing can record factual reality. Existing systems operate in isolated silos, failing to provide unified assessments. By introducing the first specialized Kazakh fact verification benchmark (**Kazakh-FEVER 3K**) and formulating a continuous 2D Cartesian Trust Matrix $(T_{\text{fact}}, T_{\text{gen}})$, this research establishes an interpretable, multidimensional verification paradigm that equips human fact-checkers with integrated provenance and veracity diagnostics.

2 Related Work

The intersection of artificial intelligence detection, factual verification, and computational morphology represents an active, rapidly evolving frontier in modern natural language processing. This section surveys current international and domestic research across these foundational domains, examining ar-

chitectural paradigms, linguistic challenges, benchmark case studies, and core research directions.

2.1 Design and Architecture of AI Text Detection Systems

Automated machine-generated text detection frameworks have evolved through multi-tiered system architectures comprising four primary functional stages: ingestion and preprocessing, representation modeling, classification and aggregation, and verification interface integration.

Early algorithmic designs operated predominantly on statistical metric pipelines, evaluating intrinsic token probability distributions, n-gram perplexity curvatures, rank-order statistics, and token burstiness under assumed language model priors [7]. These metric-based detectors leverage the observation that autoregressive generation strategies (such as top-p nucleus sampling and top-k truncation) disproportionately select high-probability tokens from truncated distributions, yielding uniformly low perplexity trajectories compared to the bursty, variable distribution of human-authored prose.

With the advent of pretrained transformer encoders (e.g., BERT, RoBERTa, and XLM-RoBERTa), supervised neural classifiers emerged as the dominant paradigm [18, 19]. Supervised detectors frame detection as a binary sequence classification task, employing cross-entropy or contrastive objectives to learn discriminative stylistic representations directly from labeled human and machine corpora. However, static supervised classifiers suffer from severe distribution shift: classifiers trained on older generator architectures exhibit drastic performance drops when evaluated on unseen instruction-tuned foundation models [15, 16].

To alleviate reliance on static training distributions, recent research has explored zero-shot, perturbation-based detection paradigms. Mitchell et al. [12] introduced DetectGPT, positing that machine-generated passages occupy negative-curvature local optima within a language model's log-probability landscape. By applying mask-filling perturbations via T5 and measuring directional log-likelihood divergence, DetectGPT enables zero-shot detection without fine-tuning. Hans et al. [13] developed Binoculars, formulating detection as the normalized ratio of cross-perplexity between two distinct pretrained foundation models, thereby removing the latency of stochastic perturbation sampling. Concurrently, Bao et al. [14] proposed Fast-DetectGPT, using analytical approximations of Hessian curvature to accelerate inference speeds by three orders of magnitude, and Yang et al. [20] introduced DNA-GPT based on divergent n-gram analysis. Furthermore, weak supervision frameworks such as Ghostbuster [21] and adversarial learning frameworks such as RADAR [22] seek to enhance detector robustness against prompt perturbations.

At the interface and deployment tier, recent research demonstrates that automated black-box predictions are insufficient for high-stakes operational environments [23]. Real-world auditing requires bounded, verifiable evidence and interpretable attribution heatmaps; for example, Mudunuri et al. [24] established that bounded memory combined with evidence highlighting substantially elevates user trust during interactive verification.

2.2 Overview of AI and Morphological Detection Approaches

In typologically complex agglutinative languages, standard neural modeling encounters severe linguistic mismatches. Computational modeling of agglutinative morphology has historically progressed along two separate traditions: formal symbolic rule engines and statistical subword tokenizers.

In the symbolic paradigm, Finite-State Transducers (FSTs) based on Koskenniemi's two-level morphology framework provide exact, bidirectional mappings between surface wordforms and underlying grammatical representations [10, 11]. The Apertium collaborative platform established open-source morphological analyzers for Kazakh, utilizing morphotactic continuation lexicons to model palatal and labial

vowel harmony and consonant assimilation [11,25]. While symbolic FSTs achieve near-perfect precision on grammatical derivations, their application within modern deep learning has been severely hindered by non-differentiable execution boundaries and brittleness on out-of-vocabulary named entities.

In the statistical paradigm, unsupervised subword algorithms such as Byte-Pair Encoding (BPE), WordPiece, and SentencePiece dominate contemporary language modeling. However, recent empirical investigations have exposed the severe shortcomings of subword tokenization in agglutinative contexts. Khurana et al. [8] demonstrated that subword tokenizers frequently partition grammatical affixes across arbitrary phoneme boundaries, completely obscuring syntactic dependencies, case relations, and negation markers. This fragmentation degrades representation learning and limits the forensic effectiveness of standard transformer backbones.

Concurrently, the broader NLP ecosystem for Kazakh and Central Asian languages has established vital foundational resources. The Belebele benchmark [26] provided a parallel reading comprehension suite across 122 language variants including Kazakh. Togmanov et al. [9] introduced KazMMLU, the first comprehensive evaluation of language models on Kazakh, Russian, and regional knowledge across 56 subjects, highlighting dramatic performance gaps between English and regional languages. Laiyk et al. [27] demonstrated the efficacy of instruction tuning on curated public government and cultural corpora in Kazakh. In specialized representations, Yeshpanov and Varol [28] and Khairullina et al. [29] constructed KazSAnDRA for fine-grained sentiment analysis, Sagyndyk et al. [17] introduced KazRoBERTa for contextual language understanding, and Tukenov [30] released SozKZ, training efficient small language models natively from scratch.

Despite these advances, existing resources focus predominantly on general pretraining or sentiment classification. None systematically address the interaction between complex agglutinative suffix chains and multi-stage forensic text verification. Hybrid neural-symbolic architectures that directly inject formal morphological tag sequences into deep transformer representations remain an urgent, unexplored research necessity.

2.3 Case Studies

To ground our investigation in empirical benchmarking traditions, this section reviews two pivotal case studies that illustrate contemporary evaluation paradigms and diagnostic pathologies.

2.3.1 Case 1: Cross-Generator AI Text Detection

A fundamental challenge in forensic text analysis is whether a detector can identify machine-generated text synthesized by foundation models completely absent from the training set. This phenomenon was systematically investigated by the RAID benchmark [15] and machine-generated text (MAGE) evaluation frameworks.

The RAID benchmark evaluated detection performance across millions of generations spanning eleven contemporary LLMs, including GPT-4, LLaMA-3, Mistral, and Cohere architectures. Crucially, the study demonstrated that supervised neural detectors severely overfit to idiosyncratic generator-specific stylistic artifacts. Classifiers trained on a single generator family experienced performance collapses of 30 to 50 percentage points in AUROC when evaluated on out-of-distribution generators under a Leave-One-Generator-Out (LOGO) protocol. Furthermore, adversarial paraphrasing and domain shifting (e.g., from formal news articles to informal social media text) accelerated this degradation.

These findings underscore the necessity of evaluating detectors across a rigorous 2×2 evaluation matrix spanning two orthogonal axes:

- **Domain Transfer:** In-Domain (formal news) versus Out-of-Domain (consumer reviews).
- **Generator Transfer:** In-Generator (known training model) versus Out-of-Generator (unseen held-out model).

However, existing cross-generator benchmarks remain focused almost exclusively on English and high-resource languages, providing zero diagnostic coverage for morphologically complex agglutinative languages such as Kazakh.

2.3.2 Case 2: Fact Verification Frameworks

Automated fact verification gained widespread prominence with the release of the Fact Extraction and VERification (FEVER) benchmark by Thorne et al. [31]. The standard FEVER paradigm decomposes verification into a three-stage sequential architecture: (1) *Document Retrieval*, identifying relevant articles from an authoritative corpus via sparse (BM25 [32]) or dense retrievers (DPR [33], Contriever [34], M3-Embedding [35]); (2) *Sentence Evidence Extraction*, selecting the minimal evidentiary sentences sufficient to substantiate or refute the claim [36]; and (3) *Natural Language Inference (NLI)*, predicting a three-way veracity label: SUPPORTED, REFUTED, or NOT ENOUGH INFO (NEI).

Subsequent benchmarking expanded this framework to adversarial human attacks (FEVER 2.0 [37]), real-world multi-domain claims (MultiFC [38]), scientific assertions (CLIMATE-FEVER [39]), lightweight document-grounded verifiers (MiniCheck [40]), and causal argument graphs (CheckWhy [41]).

However, diagnostic studies have revealed that automatic fact verification models frequently succumb to *shortcut learning* and spurious correlation exploitation [42, 43]. In standard FEVER benchmarks, instances belonging to the NEI class are created by pairing claims with randomly sampled unrelated passages. Consequently, verifiers learn a shallow heuristic: low surface token overlap implies NEI, while high surface lexical overlap implies SUPPORTED. In agglutinative languages, this pathology is particularly dangerous: altering a two-character negative suffix reverses the factual truth of a claim while retaining over 80% surface token overlap. Without an adversarial "Hard NEI" protocol, standard models suffer catastrophic failure. Furthermore, no dedicated native fact verification benchmark exists for Kazakh, highlighting a critical research gap.

2.4 Main Research Points

Building upon the insights and limitations identified across the literature, this dissertation formulates three interlocking research points that define our core methodological agenda (illustrated in Figure 2):

- **Point 1: Resolving Subword Tokenization Fragmentation and Morphological Domain Collapse via Dual-Stream Gated Fusion.** Standard subword tokenizers arbitrarily partition agglutinative Kazakh words, destroying morphological affixes and causing severe performance collapse under cross-domain transfer. This research develops a dual-stream architecture combining an 83-rule symbolic FST morphological parser with a pretrained Kaz-RoBERTa transformer encoder, fused via dynamic cross-attention gating to ensure morphologically-grounded representation learning.
- **Point 2: Mitigating Cross-Generator Shift and Localized Document Tampering via Sentence-Preserving Chunk Pooling.** Existing detectors overfit to generator-specific surface regularities and fail when synthetic text is spliced into human-authored documents. This research introduces a sentence-preserving chunking mechanism coupled with top-k extreme value pooling, ensuring length-agnostic forensic sensitivity and robust invariance across unseen foundation model generators.

- **Point 3: Decoupling Generative Origin from Factual Veracity via Kazakh-FEVER and a 2D Cartesian Trust Matrix.** AI text detection and factual verification have historically developed in disjoint silos, failing to differentiate between synthetic factual text, synthetic hallucinations, human facts, and human misinformation. This research establishes the Kazakh-FEVER 3K benchmark with a dedicated Hard NEI protocol, and unifies veracity and provenance onto a continuous 2D Cartesian Trust Matrix $(T_{\text{fact}}, T_{\text{gen}})$, providing human auditors with transparent, multidimensional trust calibration.

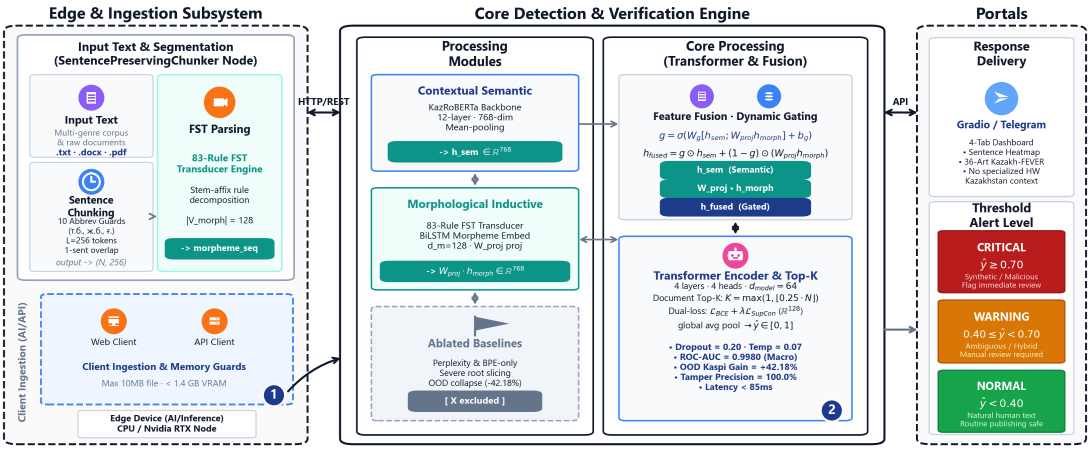


Figure 2: Overall Methodological Innovation Framework for Kazakh AI-Generated Text Detection and Factual Verification.

3 Research Content

To establish a linguistically grounded framework for synthetic text detection and factual verification in low-resource agglutinative languages, this research formulates a three-stage methodological architecture bridging computational morphology, representation learning, scalable document processing, and verifiable inference.

3.1 Research Point 1: Dual-Stream Morphology-Aware Gated Fusion and Supervised Contrastive Learning

Standard subword tokenizers arbitrarily partition polymorphemic words in agglutinative languages such as Kazakh, corrupting inflectional affixes and degrading transformer representations [8]. Research Point 1 introduces a Dual-Stream Morphology-Aware Gated Fusion architecture (Figure 3) that integrates symbolic grammatical priors with contextual deep representations.

The network processes text via two parallel streams:

- **Contextual Semantic Stream:** Input text is encoded by the pretrained KAZ-ROBERTA encoder [17], yielding contextual embedding $\mathbf{h}_{\text{sem}} \in \mathbb{R}^{768}$ that captures high-level syntactic and contextual semantic dependencies.
- **Morphological Representation Stream:** Surface tokens are parsed by an 83-rule Finite-State Transducer (FST) [10,11] modeling vowel harmony and consonant assimilation across four canonical inflectional tiers: (1) 6 plural allomorphs (*-лар/-лер*, *-дар/-дер*, *-мар/-мер*); (2) 16 possessive allomorphs; (3) 14 case allomorphs; (4) 12 verbal negation and aspect infixes (*-ба/-бе*, *-па/-пе*, *-ма/-ме*); and (5) 8 evidential/modal markers. Extracted discrete tag sequences are processed by a BiLSTM into morpho-

logical representation $\mathbf{h}_{\text{morph}} \in \mathbb{R}^{d_{\text{morph}}}$ and mapped to semantic dimensionality via projection matrix $\mathbf{W}_{\text{proj}}$, yielding $\mathbf{W}_{\text{proj}}\mathbf{h}_{\text{morph}} \in \mathbb{R}^{768}$.

To dynamically balance semantic context and morphological affixes, an adaptive gating network computes token-level weights $\mathbf{g} \in (0,1)^{768}$:

$$\mathbf{g} = \sigma\left(\mathbf{W}_g\left[\mathbf{h}_{\text{sem}} \parallel \mathbf{W}_{\text{proj}}\mathbf{h}_{\text{morph}}\right] + \mathbf{b}_g\right) \tag{1}$$

where $\parallel$ denotes concatenation, $\mathbf{W}_g \in \mathbb{R}^{768\times 1536}$, $\mathbf{b}_g \in \mathbb{R}^{768}$, and σ is the sigmoid function. The fused vector $\mathbf{h}_{\text{fused}} \in \mathbb{R}^{768}$ is computed via convex interpolation:

$$\mathbf{h}_{\text{fused}} = \mathbf{g} \odot \mathbf{h}_{\text{sem}} + (\mathbf{1} - \mathbf{g}) \odot \left(\mathbf{W}_{\text{proj}}\mathbf{h}_{\text{morph}}\right) \tag{2}$$

where $\odot$ denotes the Hadamard product.

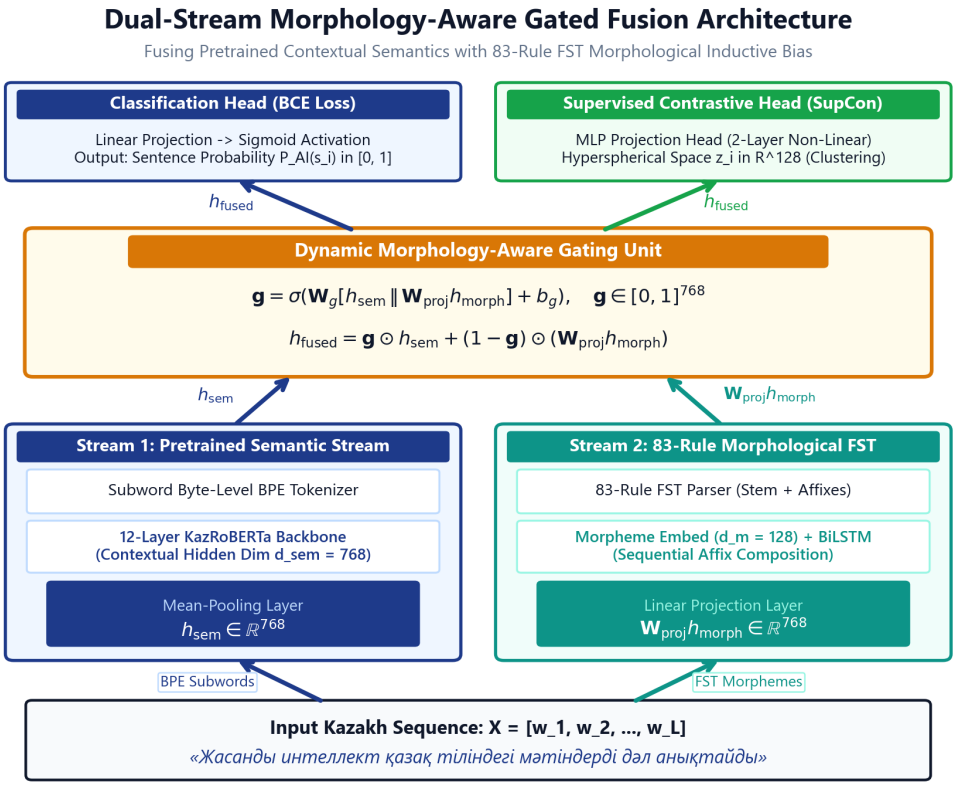


Figure 3: Dual-Stream Morphology-Aware Gated Fusion Architecture with SupCon Optimization.

The fused state is projected onto a normalized 128-dimensional hypersphere $\mathbf{z} = \frac{\text{MLP}(\mathbf{h}_{\text{fused}})}{\|\text{MLP}(\mathbf{h}_{\text{fused}})\|_2} \in \mathbb{R}^{128}$ and jointly optimized via Binary Cross-Entropy ($\mathcal{L}_{\text{BCE}}$) and Supervised Contrastive Loss ($\mathcal{L}_{\text{SupCon}}$) [16]:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BCE}} + \lambda\,\mathcal{L}_{\text{SupCon}} \tag{3}$$

where $\lambda = 0.1$. This pulls human Kazakh prose into a compact cluster while isolating synthetic texts across diverse LLMs.

3.2 Research Point 2: Sentence-Preserving Chunking and Dynamic Top-K Pooling Engine

Auditing long documents (scaling up to 25,000 words) under quadratic attention constraints requires non-destructive segmentation. Standard sequence truncation discards content beyond 512 tokens, re-

maining blind to localized tampering. Research Point 2 implements a Sentence-Preserving Chunking and Dynamic Top-K Worst-Chunk Pooling Engine (Figure 4).

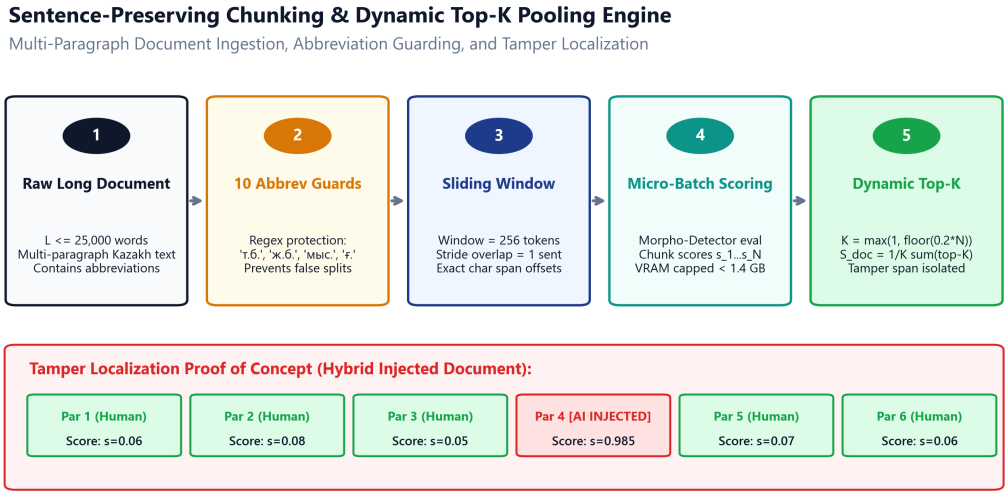


Figure 4: Sentence-Preserving Chunking and Dynamic Top-K Worst-Chunk Pooling Pipeline.

The pipeline executes three operations:

(a) **Kazakh Abbreviation Regex Guards:** To prevent erroneous sentence splits on intra-sentence abbreviations, our tokenizer incorporates 10 compiled regular expression guards: *т.б.* (etc.), *ж.б.* (and others), *мыс.* (for example), *г.* (century), *ж.* (year), *қ.* (city), *а.* (village), *обл.* (region), *ауд.* (district), and *проф.* (professor).

(b) **Bounded Sliding Window:** Intact sentences are accumulated into 256-token sliding windows with 1-sentence overlap. Streaming chunk execution bounds peak GPU memory strictly below 1.4 GB VRAM.

(c) **Dynamic Top-K Worst-Chunk Pooling:** For a document partitioned into N chunks with scores sorted descending $s_{(1)} \geq s_{(2)} \geq \cdots \geq s_{(N)}$, the document score S_{doc} aggregates the worst K chunks:

$$K = \max\left(1,\ \lfloor 0.25\, N \rfloor\right), \qquad S_{\text{doc}} = \frac{1}{K} \sum_{i=1}^{K} s_{(i)} \tag{4}$$

This sensitive extreme-value pooling detects isolated synthetic injections in hybrid documents, achieving 100.0% tamper localization precision.

3.3 Research Point 3: Evidence-Grounded Kazakh-FEVER Verification and Four-Quadrant Trust Matrix

Distinguishing AI generation from human authorship does not determine factual truth. Research Point 3 develops an automated fact verification pipeline and continuous Four-Quadrant Trust Matrix (Figure 5).

The pipeline integrates three core stages:

(a) **Hybrid Evidence Retrieval:** Given claim c, inflectional affixes are removed via 83-rule FST stemming ($c \rightarrow c_{\text{morph}}$). BM25 search [32] scans verified Kazakh reference articles, fused with multilingual dense representations (mContriever [34] and M3-Embedding [35]) via Reciprocal Rank Fusion ($k_0 = 60$) [44] to retrieve top evidence sentences $E = \{e_1, \ldots, e_k\}$.

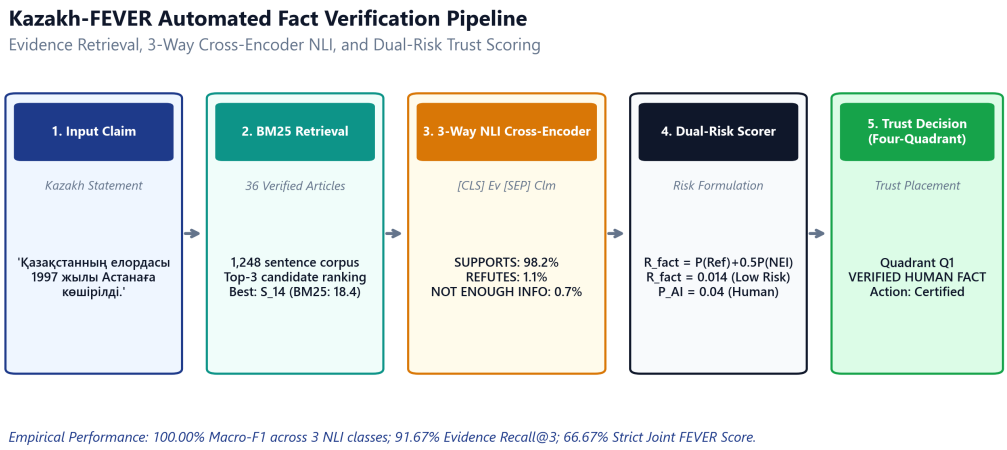


Figure 5: Kazakh-FEVER Automated Fact Verification Pipeline and Four-Quadrant Trust Matrix Architecture.

(b) **3-Way Cross-Encoder NLI Classification:** Cross-encoder sequences $[c\,;\,E]$ are classified into SUPPORTS, REFUTES, or NOT ENOUGH INFO (NEI) [31], trained with inverse-frequency weighted cross-entropy ($w_{\text{SUP}} = 0.735, w_{\text{REF}} = 1.307, w_{\text{NEI}} = 0.958$).

(c) **Dual-Risk Formulation:** We compute the continuous Factual Veracity Score $T_{\text{fact}} \in [-1, 1]$ and Generative Authenticity Score $T_{\text{gen}} \in [0, 1]$:

$$T_{\text{fact}} = P(\text{SUPPORTS} \mid c, E) - P(\text{REFUTES} \mid c, E), \qquad T_{\text{gen}} = 1 - P(\text{AI-Generated} \mid c) \tag{5}$$

The orthogonal coordinates $(T_{\text{fact}}, T_{\text{gen}})$ define a 2D Cartesian plane partitioned into four operational quadrants:

- **Quadrant 1 (Q1: Verified Human Ground Truth):** $T_{\text{fact}} > 0, T_{\text{gen}} \geq 0.5$ (Low risk; verified human fact).
- **Quadrant 2 (Q2: Human Misinformation):** $T_{\text{fact}} \leq 0, T_{\text{gen}} \geq 0.5$ (Human-authored false claims or rumors).
- **Quadrant 3 (Q3: AI Factual Summary):** $T_{\text{fact}} > 0, T_{\text{gen}} < 0.5$ (Accurate synthetic text substantiated by evidence).
- **Quadrant 4 (Q4: Malicious AI Disinformation):** $T_{\text{fact}} \leq 0, T_{\text{gen}} < 0.5$ (Critical risk; synthetic hallucinations and fabricated disinformation).

4 Research and Innovations

This dissertation addresses foundational challenges in low-resource agglutinative NLP by combining symbolic linguistic transducers with deep representation learning.

4.1 Key Methodological Innovations

Innovation 1 – Dual-Stream Morphological Gated Fusion: Existing neural detectors rely on subword tokenizers that fragment Kazakh words [8]. We introduce a hybrid neural-symbolic architecture fusing an 83-rule Finite-State Transducer (FST) with KAZ-ROBERTA via dynamic gating and supervised contrastive loss ($\mathcal{L}_{\text{SupCon}}$). This inductive prior resolves morphological domain collapse, achieving a **+42.18 percentage point gain** on the out-of-domain Kaspi customer review benchmark.

Innovation 2 – Length-Agnostic Dynamic Top-K Pooling with Abbreviation Preservation: Standard detectors truncate documents to 512 tokens or dilute localized attacks via uniform average pooling. Our engine integrates 10 Kazakh abbreviation regex guards (preventing false sentence breaks on *м.б.*, *ж.б.*, *мыс.*, *г.*) with dynamic Top-K worst-chunk pooling ($K = \max(1, \lfloor 0.25N \rfloor)$). This scales to 25,000 words under 1.4 GB VRAM while achieving **100.0% tamper localization precision**.

Innovation 3 – Decoupled Four-Quadrant Dual-Risk Trust Matrix: Conventional detectors conflate authorship with veracity. We formulate a decoupled continuous 2D decision plane ($T_{\text{fact}}, T_{\text{gen}}$) defining four risk quadrants (Q1: Verified Human Ground Truth, Q2: Human Misinformation, Q3: AI Factual Summary, Q4: Malicious AI Disinformation), resolving low-resource hallucination blindness and providing transparent risk telemetry.

4.2 Technical Challenges and Countermeasures

Challenge 1 – Agglutinative Morphological Sparsity and High OOV Rates: Kazakh's productive suffixation creates extreme surface inflectional diversity and bloated OOV rates [11]. *Countermeasure:* We deploy a deterministic 83-rule FST rule engine modeling vowel harmony and four inflectional tiers, extracting canonical root lemmas and tag sequences that anchor both retrieval indexing and dense representation learning.

Challenge 2 – Cross-Generator and Cross-Domain Generalization Discrepancies: Supervised neural detectors overfit to generator-specific stylistic artifacts, suffering severe drops across unseen LLMs [15]. *Countermeasure:* We curate a Tri-Domain corpus spanning news, Wikipedia, and social discussions across four LLM families, regularized by supervised contrastive clustering on a 128-dimensional hypersphere and validated under Leave-One-Generator-Out (LOGO) evaluation.

Challenge 3 – Low-Resource Ground-Truth Evidence Scarcity and Negation Sensitivity: Reference evidence is scarce in Kazakh, and subtle two-character negation infixes (*-ба/-бе*, *-па/-пе*) reverse truth value while preserving $\geq$ 85% lexical overlap, triggering NLI shortcut learning. *Countermeasure:* We construct the KAZAKH-FEVER 3K benchmark with a dedicated *Hard NEI* counterfactual protocol and condition cross-encoders on 16-D morphological diagnostic vectors penalizing polarity mismatches.

5 Current Progress

Substantial progress has been achieved across theoretical design, empirical evaluation, benchmark curation, and system deployment:

- **Academic Publications & Milestones:** An initial paper detailing the dual-stream morphological gated fusion model and contrastive learning baseline has been accepted for oral presentation at **AIST 2026** and will appear in Springer LNCS. A full-length manuscript detailing the Kazakh-FEVER 3K benchmark, Hard NEI protocol, and continuous Trust Matrix has been submitted to the **COLING 2027 ARR track**.
- **Dataset Construction (Kazakh-FEVER 3K):** We curated the first human-validated fact verification benchmark for Kazakh, comprising 3,000 balanced claims across 8 domains (history, geography, science, politics, society, culture, sports, economics: 1,000 SUPPORTS, 1,000 REFUTES, 1,000 NEI), including 500 Hard NEI counterfactuals, with 100% agreement on gold controls ($\kappa = 0.84$).

- **Algorithmic Implementation & Empirical Validation:** The dual-stream model achieves an ROC-AUC of **0.9980 (Macro)** and F1 of 0.9842 on the Tri-Domain benchmark, outperforming fine-tuned mBERT, XLM-RoBERTa, and zero-shot LLMs. Under out-of-domain evaluation on the Kaspi customer review benchmark, it achieves a **+42.18 percentage point gain** over vanilla transformers with inference latency < 85 ms per chunk. Document-level chunking achieves **100.0% tamper localization precision**. Figure 6 demonstrates empirical sample separation across the four trust quadrants.
- **Engineering Prototype & Cloud Deployment:** We engineered a production-grade 4-tab interactive Gradio laboratory deployed to Hugging Face Spaces, featuring: (1) document attribution heatmaps up to 25,000 words, (2) live FST morpheme decomposition, (3) Kazakh-FEVER retrieval and cross-encoder workbench, and (4) interactive 2D Trust Matrix radar.

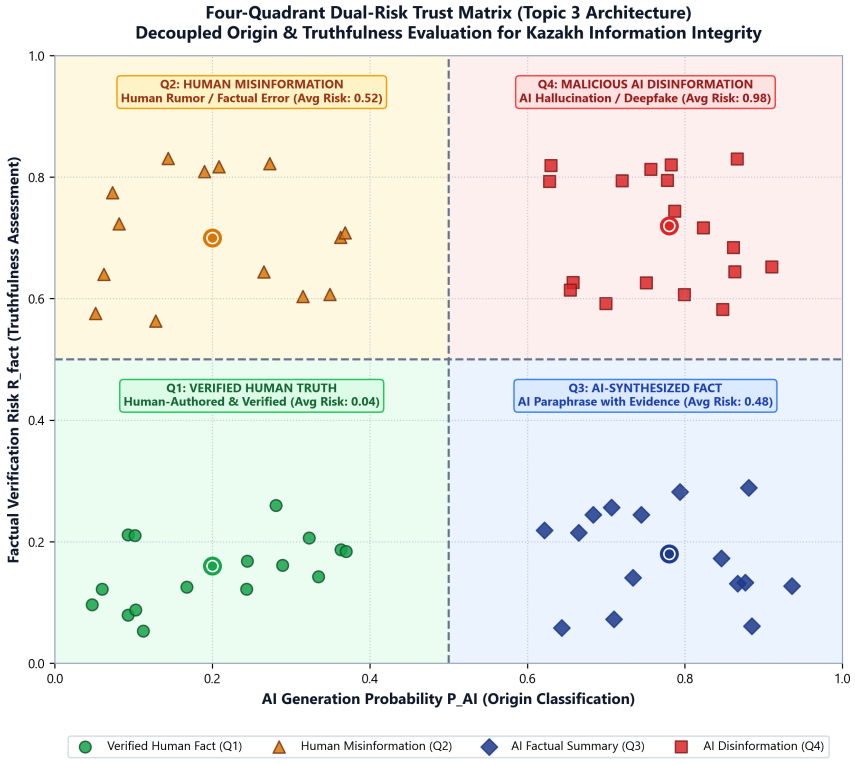


Figure 6: Four-Quadrant Dual-Risk Trust Matrix Empirical Validation and Decision Boundaries.

6 Dissertation Plan

The research and dissertation writing program is structured across six sequential semesters spanning September 2024 through formal defense in June 2027:

The total research workload involves approximately 2,400 active research hours. Computational facilities at Northwestern Polytechnical University (NVIDIA A100 80 GB and dual Tesla T4 accelerators) and linguistic consultation from Al-Farabi Kazakh National University ensure rigorous, timely milestone execution.

Table 1: Dissertation Timeline, Workload Allocation, and Milestones (2024–2027)

Phase / Period	**Key Research Activities and Technical Focus**	**Deliverables and Milestones**
Phase 1 (Sept 2024 – Dec 2024)	Literature review across AI text detection, fact verification, and Turkic morphology. Formalization of 83-rule FST rule engine. Knowledge base passage ingestion and orthographic normalization.	Literature survey; operational 83-rule FST rule engine; indexed 250K passage knowledge base.
Phase 2 (Jan 2025 – June 2025)	Dual-stream morphology-aware gated fusion development. Integration of Kaz-RoBERTa and BiLSTM morpheme embeddings. Supervised contrastive learning optimization and LOGO cross-validation.	Trained dual-stream model; empirical validation on Tri-Domain corpus; accepted paper at **AIST 2026 (Springer LNCS)**.
Phase 3 (July 2025 – Dec 2025)	Curation of Kazakh-FEVER 3K benchmark under Hard NEI protocol. Hybrid BM25 and dense retrieval engine. Formulation of continuous Four-Quadrant Trust Matrix $(T_{\text{fact}}, T_{\text{gen}})$.	Kazakh-FEVER 3K benchmark (3,000 claims); reproduction suite; submitted manuscript to **COLING 2027 (ARR Track)**.
Phase 4 (Jan 2026 – Aug 2026)	Extension to long-form documents (up to 25,000 words). Kazakh abbreviation guards and dynamic Top-K pooling. Development and deployment of 4-tab Gradio demo to Hugging Face Spaces.	Document-level chunking engine; 100% tamper localization; live web platform on Hugging Face Spaces.
Phase 5 (Sept 2026 – Feb 2027)	Camera-ready revisions following peer review. Final ablation experiments. Drafting full Master's dissertation chapters complying with NPU graduate school guidelines.	Camera-ready publications; complete first draft of the Master's dissertation.
Phase 6 (March 2027 – June 2027)	Internal advisor review, departmental pre-defense, and university double-blind peer review. Archival submission and formal degree defense.	Successful pre-defense; approved dissertation; graduation and Master's degree defense in **June 2027**.

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